EXP 1

confidentiality

gedit conf.txt
Hello World
chmod 600 conf.txt. (only owner access)
ls -l conf.txt

EXP 2

intergrity

gedit integ.txt
Hello World
md5sum integ.txt. (generate hash)
gedit integ.txt
Hello IS
md5sum integ.txt. (diff hash)

EXP 3

availability

gedit avail.txt
Hello World
chmod 000 avail.txt. (remove all access)
cat avail.txt. (permission denied)
chmod 644. avail.txt. (regain access)
cat avail.txt. (access granted)

EXP 4

conf + avail

gedit secure.txt
chmod 640 secure.txt (owner group others) (balanced)
ls -l secure.txt
cat secure.txt

Exp 5

Dac

gedit dac.txt
chmod 700 dac.txt (full control to user)
ls -l dac.txt

Exp 6

Mac

sudo apt install
sudo apt install apparmor-utils
sudo aa-status
sudo aa-complain /bin/cat (mode to log violations not block)
sudo aa-enforce. /bin/cat
cat (any restricted file)

EXP 7

rbac

```
sudo groupadd develops
sudo usermod -aG develops $USER
gedit rbac.txt
Hello World
sudo chown  :develops rbac.txt
chmod 750 rbac.txt
su - develops
ls -l rbac.txt
```

Exp 8

digital signature

```
gedit msg.txt
Hello World
openssl genrsa -out key.pem 1024
```

openssl rsa -in key.pem -pubout -out pub.pem. (extract public key)

openssl dgst -sha256 -sign key.pem -out sign.bin msg.txt. (sign)

openssl dgst -sha256 -verify pub.pem -signature sign.bin msg.txt. (verify)

exp 9

nmap

sudo apt update
sudo apt install nmap
nmap 127.0.0.1
nmap -p 22 127.0.0.1

exp 10
aes

gedit aes.txt
Hello world
cat aes.txt
openssl enc -aes-256-cbc -in aes.txt -out aes.enc. (encrypt)
cat aes.enc
openssl enc -aes-256-cbc -d -in aes.enc -out aes_dec.txt. (decrypt)
cat aes_dec.txt

exp 11

gedit rsa.txt
openssl genrsa -out key.pem 1024. (generate key)
openssl rsa -in key.pem -pubout -out pub.pem. (public key)
openssl rsautl -encrypt -pubin -inkey pub.pem -in rsa.txt -out rsa.enc. (encrypt)
cat rsa.enc

```
openssl rsautl -decrypt -inkey key.pem -in
rsa.enc -out rsa_dec.txt.     (decrypt)
cat rsa_dec.txt
```

exp 12

```
gedit data.txt
Name: Anni, Email: anni@gmail.com
Name: Rahul, Email: rahul@gmail.com
cat data.txt
sed 's/Name: [A-Za-z]*/Name: XXX/g'
data.txt > anon.txt
cat anon.txt
```